

Exercise 13

Recursive Functions: Fibonacci Sequence Generator

Introduction:

Recursion is a programming technique in which a function calls itself as part of its own definition. A recursive function solves a problem by reducing it to one or more smaller instances of the same problem, continuing until a base case is reached that can be solved directly without further calls.

The Fibonacci sequence is a well-known mathematical series in which each term is the sum of the two terms that precede it. The sequence begins with 0 and 1, producing the series 0, 1, 1, 2, 3, 5, 8, 13, and so on. Its recursive definition is expressed as follows:

$$\begin{aligned} \text{fib}(n) &= \text{fib}(n - 1) + \text{fib}(n - 2), \quad \text{for } n > 1 \\ \text{fib}(0) &= 0, \quad \text{fib}(1) = 1 \end{aligned}$$

Every recursive function must define at least one base case to prevent infinite recursion. For the Fibonacci sequence, the base cases are the zeroth term, which returns 0, and the first term, which returns 1. All other terms are resolved through the recursive case.

Learning Outcomes:

- Implement a recursive function that computes the nth term of the Fibonacci sequence.
- Apply the recursive function to generate and display a Fibonacci sequence of n terms.

Problem Description:

```
/* Determines if integer n is a non-negative integer valid as a sequence length.
   Returns true if n is greater than zero, false otherwise. */
bool isValidInput(int n);

/* Computes and returns the nth term of the Fibonacci sequence using recursion.
   Returns 0 if n is 0, returns 1 if n is 1, and returns the sum of the two
   preceding terms for all n greater than 1. */
int fibonacci(int n);

/* Prints the first n terms of the Fibonacci sequence to standard output.
   Uses the function fibonacci. */
void printSequence(int n);

/* Reads a positive integer from the user and returns it.
   Continues to prompt until a valid value is entered. Uses the function isValidInput
int getInput();
```

Call all the functions created in function main().